



## AQ10.4: Activity Questions 4 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

Let  $f(x, y)$  be defined in a neighbourhood of  $(a, b)$ . When is  $(a, b)$  a critical point of  $f$ ?

☐

$$f(a, b) = 0$$

☐

$$f_x(a, b) = 0$$

☐

$$f_x(a, b) = 0 = f_y(a, b)$$

☐

$f_x$  does not exist.

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

$$f_x(a, b) = 0 = f_y(a, b)$$

$f_x$  does not exist.

1 point

Consider  $f(x, y) = \sqrt{5x^2 + 3y + 2}$

$\sqrt{\quad}$

.

☐

Any point on the line  $x + y = 1$  in the domain of definition of  $f$  satisfies  $f_y = 0$ .

☐

There is no  $(x, y)$  such that  $f_y = 0$ .

☐

Solutions of  $5x^2 + 3y + 2 = 0$  are critical points of  $f$ .

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

There is no  $(x, y)$  such that  $f_y = 0$ .

Solutions of  $5x^2 + 3y + 2 = 0$  are critical points of  $f$ .

1 point

A critical point

☐

is a point of maximum.

☐

may be a saddle point.

☐

may be a point of local minimum.

☐

can be obtained by equating  $f_x$  and  $f_y$  to 0.

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

may be a saddle point.

may be a point of local minimum.



can be obtained by equating  $f_x = 0$  and  $f_y = 0$  to 0.

1 point

Let  $f(x, y) = x^2 y$ . Then

☐

$(0, 0)$  is a critical point.

☐

$(0, y)$  for  $y \in \mathbb{R}$  are critical points.

☐

$f_x > 0$  at the critical points.

☐

$f_y = 0$  at the critical points.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(0, 0)$  is a critical point.

$(0, y)$  for  $y \in \mathbb{R}$  are critical points.

$f_y = 0$  at the critical points.

1 point

For  $f(x, y) = x^3 + 2xy - 2x - 4y$ , which of the following is/are true?

☐

$(2, -5)$  is a critical point.

☐

$(2, 5)$  is a critical point.

☐

$f_y = 0$  is a straight line parallel to the  $y$ -axis.

☐

The system of equations  $f_x = 0$ ,  $f_y = 0$  has a unique solution.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(2, -5)$  is a critical point.

$f_y = 0$  is a straight line parallel to the  $y$ -axis.

The system of equations  $f_x = 0$ ,  $f_y = 0$  has a unique solution.

1 point

Choose the correct statement(s) about the function  $f(x, y) = xy^2 + 2x$ .

☐

$f_x = 0$  only for 2 values of  $x$ .

☐

$f_y = 0$  along both  $x$  and  $y$  axes.

☐

There are no critical points of  $f$ .

☐

$(0, 0)$  is a point of local minimum.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$f_y = 0$  along both  $x$  and  $y$  axes.

There are no critical points of  $f$ .

1 point

For  $f(x, y) = \frac{x^3}{3} + y^2 + 2xy - 6x - 3y + 4$   $f(x, y) = 3x^3 + y^2 + 2xy - 6x - 3y + 4$

☐

$f_y = 0$   $f_y = 0$  is a straight line passing through the origin.

☐

$(-1, \frac{5}{2})$   $(-1, 25)$  and  $(3, -\frac{3}{2})$   $(3, -23)$  are critical points.

☐

$f_x(-1, \frac{5}{2}) = 0$   $f_x(-1, 25) = 0$ .

☐

$f_y(3, -\frac{3}{2}) = 2$   $f_y(3, -23) = 2$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(-1, \frac{5}{2})$   $(-1, 25)$  and  $(3, -\frac{3}{2})$   $(3, -23)$  are critical points.

$f_x(-1, \frac{5}{2}) = 0$   $f_x(-1, 25) = 0$ .

1 point

Choose the correct statement(s) about  $f(x, y) = x^4 + y^4$   $f(x, y) = x^4 + y^4$ .

☐

There are no critical points.

☐

There is exactly one critical point.

☐

$(0, 0)$   $(0, 0)$  is a point of global minimum.

☐

$(0, 0)$   $(0, 0)$  is a point of local minimum.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

There is exactly one critical point.

$(0, 0)$   $(0, 0)$  is a point of global minimum.

$(0, 0)$   $(0, 0)$  is a point of local minimum.

**Level 2**

1 point

Let  $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$   $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$ . Then

☐

$(3, 1)$   $(3, 1)$  is a critical point.

☐

$(2, 1)$   $(2, 1)$  is a critical point.

☐

Value of  $f$  at  $(2, 1)$   $(2, 1)$  is -4.

☐

Value of  $f$  at  $(3, 1)$   $(3, 1)$  is -9.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(3, 1)$   $(3, 1)$  is a critical point.

Value of  $f$  at  $(2, 1)$   $(2, 1)$  is -4.

1 point

Let  $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$   $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$  be defined on the box  $0 \leq x \leq 20$   $0 \leq x \leq 2$  and  $0 \leq y \leq 10$   $0 \leq y \leq 1$ . Then which of the following options is (are) true?

☐

Along the line joining  $(0, 0)$  and  $(2, 0)$ , the minimum is attained at  $(2, 0)$ .

☐

$(0, 1)$  is a point of extremum along the line joining  $(0, 1)$  and  $(2, 1)$ .

☐

$f(0, 1) = 1$

☐ None of the above.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Along the line joining  $(0, 0)$  and  $(2, 0)$ , the minimum is attained at  $(2, 0)$ .

$(0, 1)$  is a point of extremum along the line joining  $(0, 1)$  and  $(2, 1)$ .

1 point

Let  $f(x, y) = \sin x \cos y$ . Then

☐

$((2k+1)\frac{\pi}{2}, k\pi)$  for  $k \in \mathbb{N}$  are critical points of  $f$ .

☐

$((2k+1)\frac{\pi}{2}, (2k+1)\frac{\pi}{2})$  for  $k \in \mathbb{N}$  are critical points of  $f$ .

☐

$(k\pi, k\pi)$  for  $k \in \mathbb{N}$  are critical points of  $f$ .

☐

$(k\pi, (2k+1)\frac{\pi}{2})$  for  $k \in \mathbb{N}$  are critical points of  $f$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$((2k+1)\frac{\pi}{2}, k\pi)$  for  $k \in \mathbb{N}$  are critical points of  $f$ .

$(k\pi, (2k+1)\frac{\pi}{2})$  for  $k \in \mathbb{N}$  are critical points of  $f$ .

1 point

Let  $f(x, y) = e^{-x^2 + y - \frac{y^3}{3}}$ . Then

☐

$f_x(1, 0) = -\frac{2}{e}f_x(1, 0) = -e$ .

☐

$f_x(0, 0) = 1$

☐

$f_y(0, -1) = 0$

☐

$f_y(1, 0) = e$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$f_x(1, 0) = -\frac{2}{e}f_x(1, 0) = -e$ .

$f_y(0, -1) = 0$

1 point

Let  $f(x, y) = e^{-x^2 + y - \frac{y^3}{3}}$ . Then

☐

$(0, 1)$  is a critical point.

☐

$(0, -1)$  is a critical point.

☐

(1, 1)(1, 1) is a critical point.

☐

(1, - 1)(1, -1) is a critical point.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(0, 1)(0, 1) is a critical point.

(0, - 1)(0, -1) is a critical point.

*1 point*

Find three positive numbers whose sum is 12 such that the sum of their squares is minimum. The mathematical expression that best defines this problem is

☐

minimize  $x^2 + y^2 + z^2$  such that  $x + y + z = 12$ ,  $x, y, z > 0$ .

☐

minimize  $x + y + z$  such that  $x^2 + y^2 + z^2 = 12$ ,  $x, y, z \geq 0$ .

☐

minimize  $x^2 + y^2 + z^2$  such that  $x + y + z = 12$ ,  $x, y, z \geq 0$ .

☐

minimize  $x + y + z$  such that  $x^2 + y^2 + z^2 = 12$ ,  $x, y, z \geq 0$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

minimize  $x^2 + y^2 + z^2$  such that  $x + y + z = 12$ ,  $x, y, z > 0$ .

[Check Answers](#)

Your score is: 0/14